

SAI BHARGAVI PUSULURI

Data Engineer

☎ +1(216) 438-0388 ✉ saibhargavipusuluri99@gmail.com 🌐 <https://www.linkedin.com/in/sai-p-121b5a222>

OBJECTIVE

- Data Engineer with 4+ years of experience in designing and implementing scalable data pipelines, ETL workflows, and real-time data processing solutions. Proficient in cloud platforms **Azure, AWS, GCP**, big data technologies **Apache Spark, Kafka, Hadoop**, and programming languages **Python, SQL, Java, Scala**. Adept at leveraging modern data architectures, data lakes, and warehouses **Snowflake, Redshift, Synapse** to drive actionable insights. Skilled in CI/CD, infrastructure as code **Terraform, CloudFormation**, and data visualization **Power BI, Tableau**. Seeking to leverage expertise in data engineering, analytics, and cloud solutions to optimize business intelligence and decision-making in a dynamic, data-driven environment.

PROFILE SUMMARY

- **Experienced Data Engineer** with 4+ years of expertise in **data pipeline design, ETL development, and big data solutions** across **Azure, AWS, and GCP**.
- Proficient in **Python, SQL, Scala, and Java** for **data processing, transformation, and analysis**.
- Strong expertise in **big data technologies** such as **Apache Spark, Hadoop, Kafka, Snowflake, and Databricks** for handling **large-scale datasets**.
- Skilled in **data warehousing and data lakes** using **Azure Synapse, AWS Redshift, and Google BigQuery**.
- Hands-on experience in **real-time data streaming** with **Kafka, AWS Kinesis, and Google Pub/Sub**.
- Expertise in **ETL tools** such as **Azure Data Factory, AWS Glue, Apache NiFi, Talend, and Informatica** for **seamless data integration**.
- Proficient in **CI/CD, DevOps, and automation** using **Terraform, Jenkins, Azure DevOps, and GitHub Actions**.
- Strong knowledge of **cloud-based data processing** with **AWS Lambda, Azure Functions, and GCP Dataflow**.
- Experienced in **data visualization and reporting** using **Power BI, Tableau, and Looker**.
- Skilled in **data security, governance, and performance optimization** across **PostgreSQL, MySQL, Oracle, and MongoDB**.
- Adept at integrating **REST APIs, GraphQL, and third-party systems** for **seamless data exchange**.
- Experience in **AI/ML model deployment** using **Azure ML, AWS SageMaker, and TensorFlow** for **predictive analytics**.
- Proven ability to **collaborate with cross-functional teams** to deliver **scalable, high-performance, and cost-effective data solutions**.

Education

Kent State University

Masters in Computer Science

Kent, USA

Certifications

- Google Cloud Certified – Associate Cloud Engineer
- Microsoft Certified - Azure Data Engineer - Associate
- AWS Certified: Data Engineer – Associate

Experience

Citizens Bank

April 2024 – Present

Azure Data Engineer

Providence, Rhode Island

- Designed and implemented scalable ETL pipelines using **Azure Data Factory**, optimizing data ingestion from diverse financial systems.
- Developed and maintained real-time data streaming solutions using **Apache Kafka** and **Spark Streaming** for fraud detection and transaction monitoring.
- Leveraged **Azure Databricks** and **Synapse Analytics** to transform and analyze structured and semi-structured banking data, improving reporting accuracy.
- Integrated **Azure Logic Apps** and **Azure Functions** to automate workflows, reducing manual processing efforts by 40%.
- Implemented **Azure Purview** for enterprise-wide data governance and compliance with financial regulations.
- Developed **Power BI** dashboards to visualize key banking metrics, integrating with **Azure Analysis Services** for efficient data retrieval.
- Utilized **Delta Lake** for managing and processing historical banking data, enabling faster analytics and improved query performance.
- Implemented advanced security measures using **Azure Key Vault** and **Managed Identities** to protect sensitive customer data.
- Automated deployment pipelines using **Terraform** and **Azure DevOps**, streamlining infrastructure provisioning and reducing manual errors.
- Designed and optimized **T-SQL** stored procedures and queries to enhance database performance and minimize execution time.
- Developed machine learning pipelines in **Azure ML** for credit risk assessment, integrating predictive models into banking applications.
- Ensured compliance with **GDPR** and financial industry regulations through data masking, encryption, and access control mechanisms.
- Orchestrated data workflows using **Apache Airflow** and **Azure Data Factory triggers** to ensure smooth batch and real-time data processing.
- Conducted root cause analysis and troubleshooting for data quality issues, leveraging **Azure Monitor** and **Application Insights** for proactive resolution.

FM Global's Johnston

Nov 2023 – March 2024

AWS Data Engineer

Rhode Island, USA

- Architected and optimized AWS-based data processing solutions, leveraging **AWS Glue**, **Redshift**, **EMR**, and **Kinesis** to improve data ingestion and transformation efficiency.
- Designed scalable **ETL pipelines** using **Apache Airflow**, **AWS Lambda**, and **Apache Spark**, enabling automated, high-performance data workflows.
- Implemented **real-time data streaming pipelines** utilizing **Apache Kafka**, **AWS Kinesis**, and **DynamoDB Streams**, ensuring seamless integration with business intelligence systems.
- Developed **serverless data transformation solutions** using **AWS Lambda**, **Step Functions**, and **S3**, reducing infrastructure costs and improving scalability.
- Integrated **AWS Data Lake**, **Snowflake**, and **Redshift** for structured and unstructured data processing, enhancing data analytics capabilities.
- Spearheaded **CI/CD implementation** using **GitHub Actions**, **Jenkins**, and **Terraform**, streamlining deployment automation for data engineering workflows.
- Designed **event-driven data pipelines** using **SNS**, **SQS**, and **AWS Step Functions**, improving system reliability and response times for business-critical applications.
- Managed **cross-platform data integration** by leveraging **AWS Glue** and **Apache NiFi**, facilitating seamless data movement across diverse cloud and on-premises sources.

- Engineered **distributed data processing workflows** using **Hadoop, Spark, and Flink**, improving computational efficiency for large-scale data analytics.
- Developed and optimized **SQL queries and stored procedures** in **PostgreSQL and MySQL**, enhancing query performance and data retrieval speed.

CARE Health Insurance

July 2022 – July 2023

GCP Data Engineer

Mumbai, India

- Designed and optimized **scalable data pipelines** using **Google Cloud Dataflow** and **Apache Beam** for efficient data ingestion and transformation across multiple healthcare data sources.
- Implemented **real-time data streaming solutions** leveraging **Google Pub/Sub** and **Cloud Dataproc**, ensuring near-instant processing of healthcare transactions and claims data.
- Developed **automated ETL workflows** using **Apache NiFi, Talend, and Google Cloud Dataflow**, improving data consistency and integration across various systems.
- Designed **data models and warehouses** using **Google BigQuery, Cloud SQL, and Cloud Spanner** to facilitate efficient querying, reporting, and analytics.
- Utilized **Apache Kafka** and **Flume** to handle large-scale, event-driven healthcare data ingestion with minimal latency.
- Optimized **data storage and retrieval** for structured and unstructured data, leveraging **Hadoop ecosystem** components such as **HDFS, Hive, and Pig** to enhance query performance.
- Implemented **CI/CD pipelines** using **Git, Maven, Jenkins, and Cloud Build** to automate deployments and maintain efficient data operations.
- Developed **predictive analytics solutions** using **PySpark** and **Snowflake**, enabling advanced decision-making in healthcare insurance claim processing.
- Ensured **data security and compliance** by implementing **access control policies, encryption mechanisms, and auditing** using **IAM roles** and **Google Cloud Security best practices**.
- Designed **serverless data processing solutions** using **Cloud Functions** to automate event-driven tasks and reduce infrastructure overhead.

Aurobindo Pharma

Feb 2021 – June 2022

Data Engineer

Mumbai, India

- Designed and optimized **high-performance ETL pipelines** using **Apache NiFi, Talend, and AWS Glue**, ensuring seamless data integration from multiple pharmaceutical systems into **Redshift, Snowflake, and PostgreSQL**.
- Implemented **big data processing workflows** utilizing **Apache Spark, Hadoop, and PySpark**, improving pharmaceutical supply chain analytics by 40%.
- Developed and deployed **real-time streaming solutions** using **Apache Kafka, AWS Kinesis, and Flume**, enhancing live tracking of drug inventory and production.
- Leveraged **Azure Synapse Analytics** and **Amazon Redshift** to manage large-scale **pharmaceutical datasets**, optimizing SQL queries and improving data retrieval speeds.
- Engineered **serverless data workflows** using **AWS Lambda and DynamoDB**, automating data ingestion and reducing operational overhead.
- Implemented **machine learning models** with **TensorFlow, scikit-learn, and Azure ML**, forecasting drug demand and optimizing manufacturing processes.
- Designed and secured **data lakes** using **AWS S3 and Azure Data Lake**, ensuring compliance with pharmaceutical industry regulations.
- Built interactive **BI dashboards** in **Power BI and Tableau**, providing real-time insights into drug production, distribution, and regulatory compliance.
- Integrated **CI/CD pipelines (Jenkins, GitLab, Azure DevOps)** for automated testing and deployment of data engineering solutions.
- Automated **infrastructure provisioning** using **Terraform and CloudFormation**, enabling efficient resource management and scalability.

Technical Skills

Cloud Platforms: Azure (Azure Data Factory, Synapse Analytics, Databricks, Data Lake), AWS (Glue, Redshift, EMR, Kinesis, Lambda), GCP (Big Query, Dataflow, Pub/Sub, Dataproc).

Big Data Streaming: Apache Spark, Hadoop (HDFS, MapReduce, YARN, Hive, Pig, HBase), Apache Kafka, AWS Kinesis, Google Pub/Sub, Apache Flume.

ETL Data Pipelines: Informatica, Talend, SSIS, Apache NiFi, Azure Data Factory, AWS Glue, Dataflow.

Programming Languages: Java, Python (Pandas, PySpark), Scala, SQL, Shell Scripting (Bash).

Machine Learning AI: TensorFlow, scikit-learn, AWS SageMaker, Azure Machine Learning.

Databases Data Warehousing: Oracle, MySQL, PostgreSQL, MongoDB, Snowflake, Amazon Redshift, Google Cloud SQL, Cloud Spanner.

Workflow Orchestration: Apache Airflow, Oozie, Azure Data Factory Triggers.

Data Visualization BI Tools: Tableau, Power BI, Looker.

Infrastructure as Code (IaC): Terraform, Bicep, AWS CloudFormation.

Version Control CI/CD: Git, GitHub Actions, Jenkins, Azure DevOps, AWS CodePipeline.

Security Monitoring: Azure Monitor, AWS CloudWatch, GCP Operations Suite, Kerberos.